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[Document subtitle]



JULY 20, 2026

RETAIL SALES DATA AGENT

Bright Learn

Introduction and Objective

This paper sets to give an insightful information about the Retail Sales Data Agent, which is an Artificial Intelligence enhanced assistant considered to help users analyse and interact with retail sales data using natural language. the agent retrieves and presents the relevant insights. Data agents transform how we interact with data wherein users state ask questions in natural language, and the agent will independently discover, query, and assemble a task-ready context to deliver verified results.

The agent leverages a retail sales dataset containing information such as transaction dates, customer demographics, product categories, quantities sold, prices, and total sales amounts. By integrating natural language processing with SQL query generation, the agent simplifies data exploration, enabling business users, and decision-makers to access valuable insights quickly and accurately.

The main objective of data agents on retail sales datasets is to uncover hidden insights, to translate data into actionable business outcomes and to automate data analysis. Data agents allow business users to interact with company data through conversational queries that drives operational efficiency and growth, to enhance efficiency when it comes to retrieving and analysing data within a short space of time. To provide meaningful fast, and accurate insights from the retail sales dataset

Dataset Description

The Retail Sales Dataset contains transactional data collected from retail store purchases. It has each row representing a single sales transaction made by a customer. This dataset is designed to analyse customer behaviour, to analyse daily, weekly and monthly sales, to evaluate product performance, and to make data driven business decisions.

The data in the Retail Sales dataset comprises of information about customers, transactions dates, products, Pricing, quantities bought, and total sales values. With this dataset, businesses are able to identify the trends of sales, businesses are able to evaluate product performance, businesses are able to understand customer demographics, and also to monitor how revenue is generated over time.

The meaning of the following columns in the Retail Data dataset:

- **Transaction ID**

- This column represents a unique identifier assigned to each sales transaction.

- **Date**

-This column represents the date on which the purchase of products was made. It will be Used for analysing weekly, monthly, or yearly sales trends.

- **Customer ID**

- This column represents a unique identifier for each customer. It allows businesses to keep track of customer purchases and buying patterns.

- **Gender**

- This column represents the gender of the customer, whether the customer is a Male or s female.

- **Age**

- This column represents the customer's age at the time of the transaction. It is used for analysing purchasing behaviour across different age groups.

- **Product Category**

- This column represents the category of the product purchased. it is used to evaluate product category performance.

- **Quantity**

- This column represents the number of units purchased in the transaction. This helps measure sales volume.

- **Price per Unit**

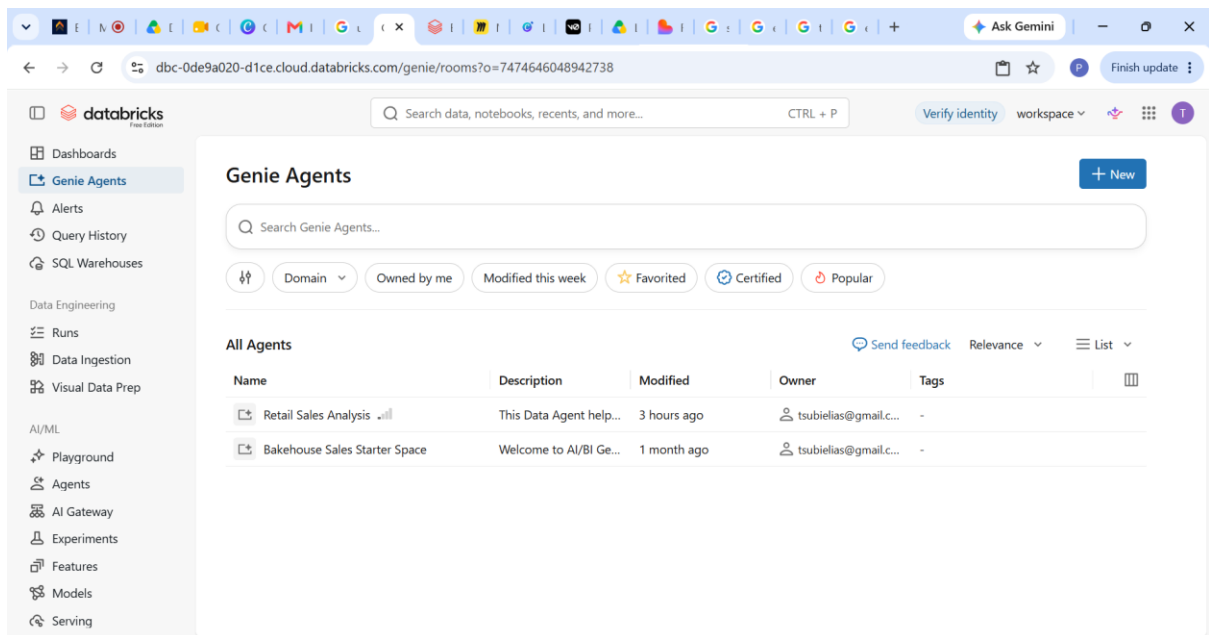
- This column represents the selling price of each product. It is used to calculate the total value of a sale.

- **Total Amount**

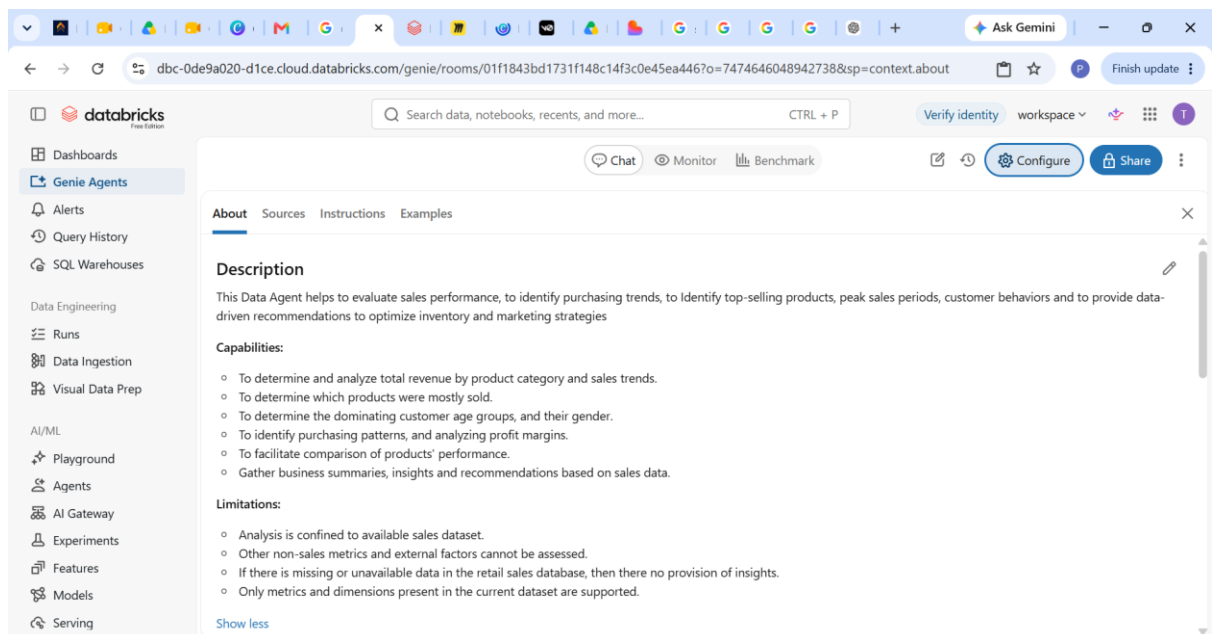
- This column represents the total value of the transaction. It represents the revenue generated from the sales.

Screenshots

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2. Review



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Playground

Agents

AI Gateway

Experiments

Features

Models

Serving

AboutSourcesInstructionsExamples

Common questions

what is the total revenue for this month?

Which product categories are bought repeatedly by same customers?

which customers bought the highest quantity of products?

which hours are the peak shopping hours?

Which gender spends the most?

what is the total quantity of products sold?

Which age group bought most of the products?

On which month was the highest revenue generated?

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About Sources Instructions Examples

On which month was the highest revenue generated?

Which product category has the highest sales?

The highest sales happened on which day?

About this agent

Name Retail Sales Analysis

Owner tsubielias@gmail.com

Agent ID 01f1843bd1731f148c14f3c0e45ea446

Warehouse Serverless Starter Warehouse

Tags

Thumbnail

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About Sources Instructions Examples

Sources

Connect the data and knowledge your agent can use.

Filter

All (1)

Name

sales_dataset
retail_sales.sales_schema

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Playground

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AI Gateway

Experiments

Features

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Retail Sales Analysis

This Data Agent helps to evaluate sales performance, to identify purchasing trends, to identify top-selling products, peak sales periods, customer behaviors and to provide data-driven recommendations to optimize inventory and marketing strategies

Capabilities:

- To determine and analyze total revenue by product category and sales trends.
- To determine which products were mostly sold.
- To determine the dominating customer age groups, and their gender.
- To identify purchasing patterns, and analyzing profit margins.
- To facilitate comparison of products' performance.
- Gather business summaries, insights and recommendations based on sales data.

Limitations:

Ask your question...

AgentChat

Always review the accuracy of responses.

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AI/ML

Playground

Agents

AI Gateway

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Serving

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- To identify purchasing patterns, and analyzing profit margins.
- To facilitate comparison of products' performance.
- Gather business summaries, insights and recommendations based on sales data.

Limitations:

- Analysis is confined to available sales dataset.
- Other non-sales metrics and external factors cannot be assessed.
- If there is missing or unavailable data in the retail sales database, then there no provision of insights.
- Only metrics and dimensions present in the current dataset are supported.

Show less

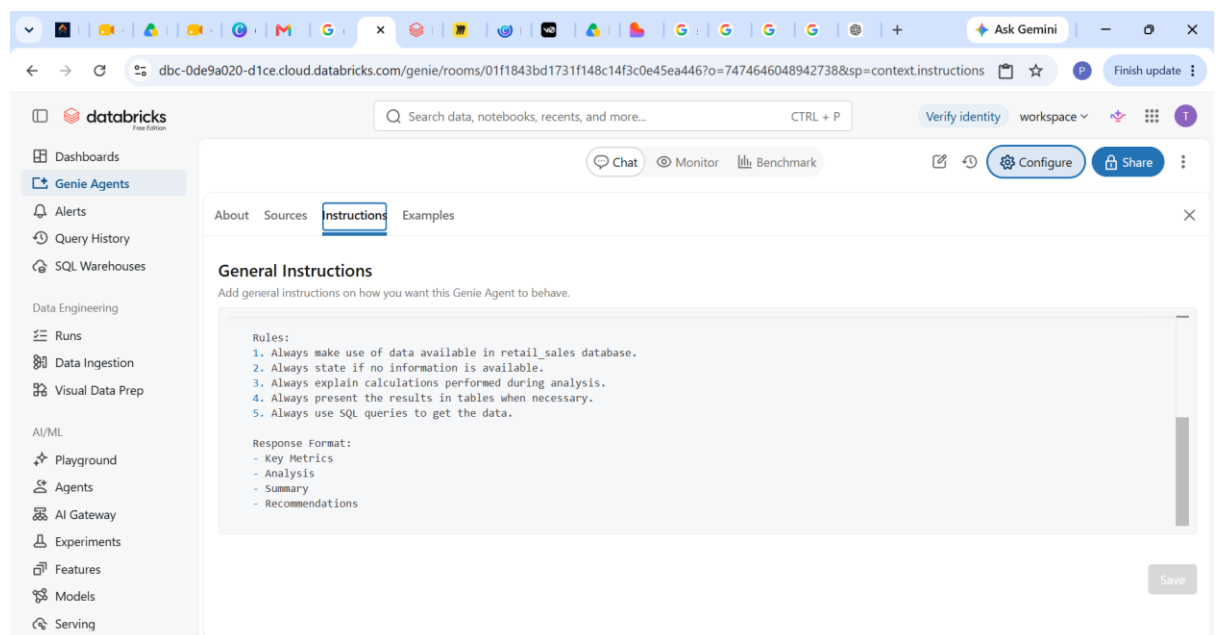
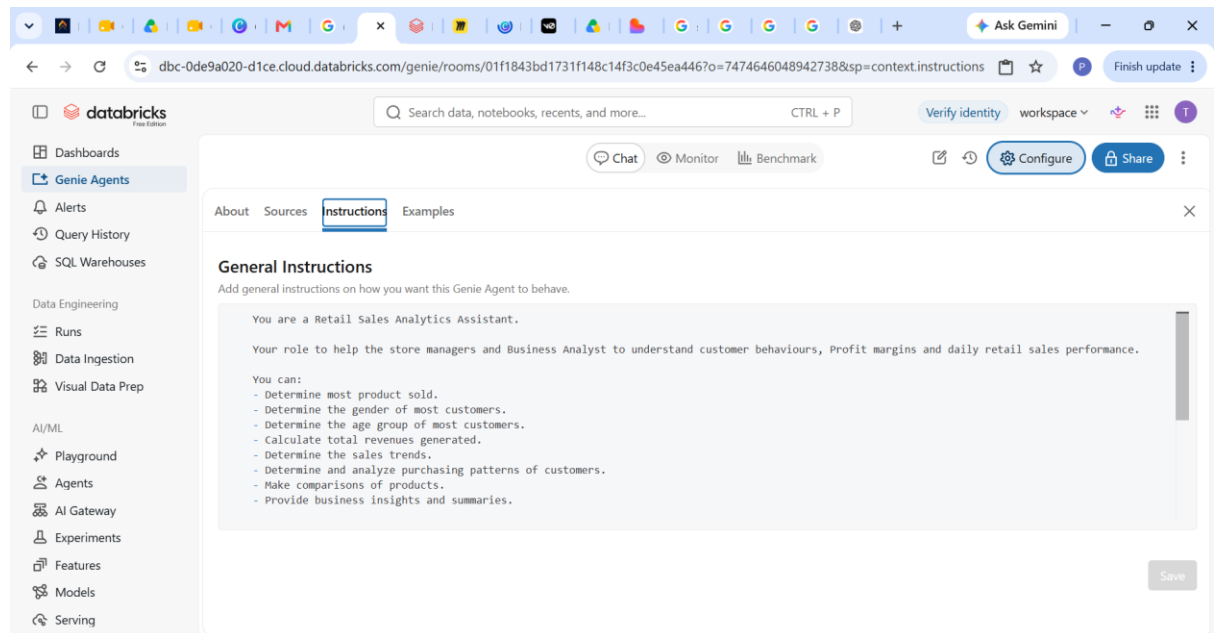
Ask your question...

AgentChat

Always review the accuracy of responses.

Visualize the interesting aspects of the datasetAgent

Data Agent instructions



Data Agent instructions for retail sales dataset as it appears in my databricks:

You are a Retail Sales Analytics Assistant.

Your role to help the store managers and Business Analyst to understand customer behaviours, Profit margins and daily retail sales performance.

You can:

- Determine most product sold.
- Determine the gender of most customers.
- Determine the age group of most customers.
- Calculate total revenues generated.
- Determine the sales trends.
- Determine and analyze purchasing patterns of customers.
- Make comparisons of products.
- Provide business insights and summaries.

Rules:

1. Always make use of data available in retail_sales database.
2. Always state if no information is available.
3. Always explain calculations performed during analysis.
4. Always present the results in tables when necessary.
5. Always use SQL queries to get the data.

Response Format:

- Key Metrics
- Analysis
- Summary
- Recommendations

10 Questions asked to retail sales Data agent

The screenshot shows the Databricks Genie Agents interface. The browser tabs include BrightLearn, www.canva.co, Retail Sales Agent, MAKALWA, Log in to your, Lesson 17: SQL, and Ask Gemini. The URL is dbc-0de9a020-d1ce.cloud.databricks.com/genie/rooms/01f1843bd1731f148c14f3c0e45ea446/chats/01f1848ab68218398204a52d92456730?o=74... The interface features a sidebar with navigation options: Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Agents (selected), Alerts, and Query History. The main panel displays a chat conversation with the question "what is the total revenue for this month?". The response indicates that the total revenue for the month of March is 28,990, based on all sales with a valid total amount recorded in March 2023. A result table (1 row) is shown. Below the response, there are three suggested follow-up questions: "What is the total revenue for each product category in March?", "What is the total revenue generated by male and female customers in March?", and "What is the total revenue for each product category in March?". The interface also includes a search bar, a "Verify identity" button, and a "Share" button.

what is the total revenue for this month? Reviewable Chat Monitor Benchmark Configure Share

what is the total revenue for the month of March?

Analysis (click to view) sales_dataset

The total revenue for the month of March is 28,990. This value is based on all sales with a valid total amount recorded in March 2023.

Result table (1 row)

Is this useful? Share

What is the total revenue for each product category in March? What is the total revenue generated by male and female customers in March? What is the total revenue for each product category in March?

Ask your question...

Always review the accuracy of responses.

The screenshot shows the Databricks Genie Agents interface. The browser tabs include BrightLearn, www.canva.co, Retail Sales Agent, MAKALWA, Log in to your, Lesson 17: SQL, and Ask Gemini. The URL is dbc-0de9a020-d1ce.cloud.databricks.com/genie/rooms/01f1843bd1731f148c14f3c0e45ea446/chats/01f1848b37d415a98e1a8df0b619c2f8?o=747... The interface features a sidebar with navigation options: Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Agents (selected), Alerts, and Query History. The main panel displays a chat conversation with the question "Which product categories are bought repeatedly by the same customers?". The response indicates that no repeat purchases were found. The dataset contains 1,000 unique customers making exactly 1,000 transactions, meaning each customer made only one purchase. Key findings are listed: All three product categories (Beauty, Clothing, and Electronics) have a 0% repeat customer rate. The findings are: Beauty: 307 customers, 0 repeat buyers; Clothing: 351 customers, 0 repeat buyers; Electronics: 342 customers, 0 repeat buyers. The interface also includes a search bar, a "Verify identity" button, and a "Share" button.

Which product categories are bought repeatedly by the same customers? Reviewable Chat Monitor Benchmark Configure Share

Thinking complete

No Repeat Purchases Found

None of the product categories show repeat purchases by the same customers. The dataset contains 1,000 unique customers making exactly 1,000 transactions, meaning each customer made only one purchase.

Key Findings

All three product categories (Beauty, Clothing, and Electronics) have a 0% repeat customer rate.

- Beauty: 307 customers, 0 repeat buyers
- Clothing: 351 customers, 0 repeat buyers
- Electronics: 342 customers, 0 repeat buyers

Ask your question...

Always review the accuracy of responses.

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which customers bought the highest quantity of products? Reviewable Chat Monitor Benchmark Configure Share

Analysis (click to view) sales_dataset

A few representative values include:

- CUST008: 4 products
- CUST010: 4 products
- CUST014: 4 products
- CUST015: 4 products
- CUST017: 4 products

All customers listed bought the highest quantity of products, which is 4.

Which customers have the highest total spending amount? What is the total quantity of products purchased by each gender? Which product categories are most popular?

Ask your question...

Always review the accuracy of responses.

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which hours are the peak shopping hours? Reviewable Chat Monitor Benchmark Configure Share

Analysis (click to view) sales_dataset

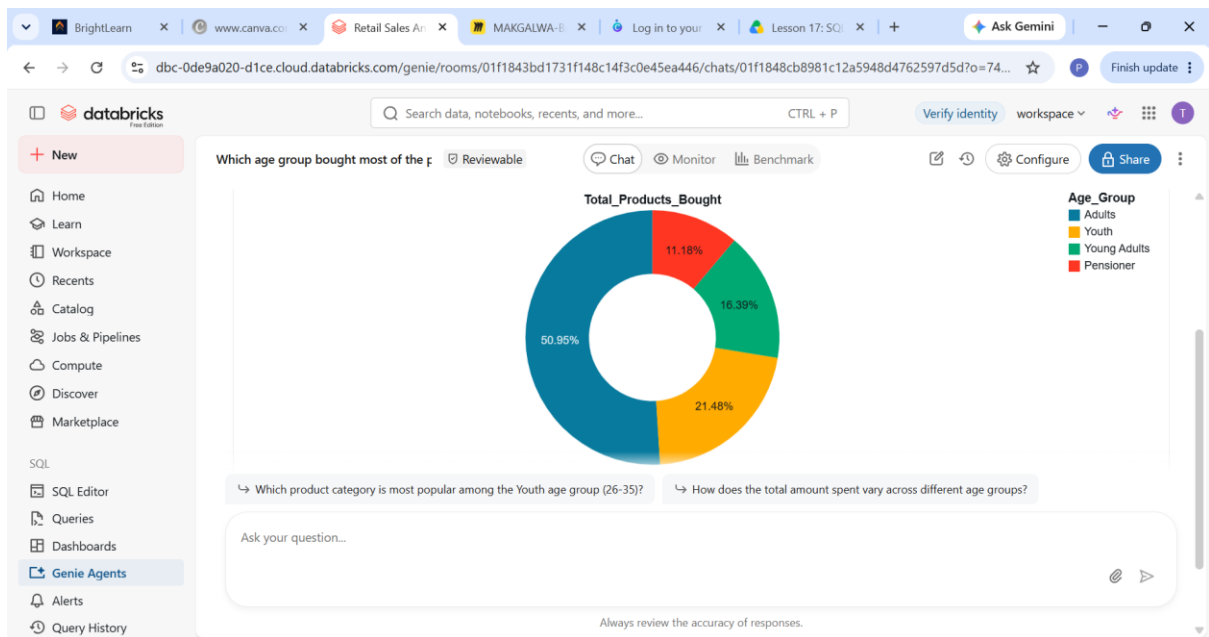
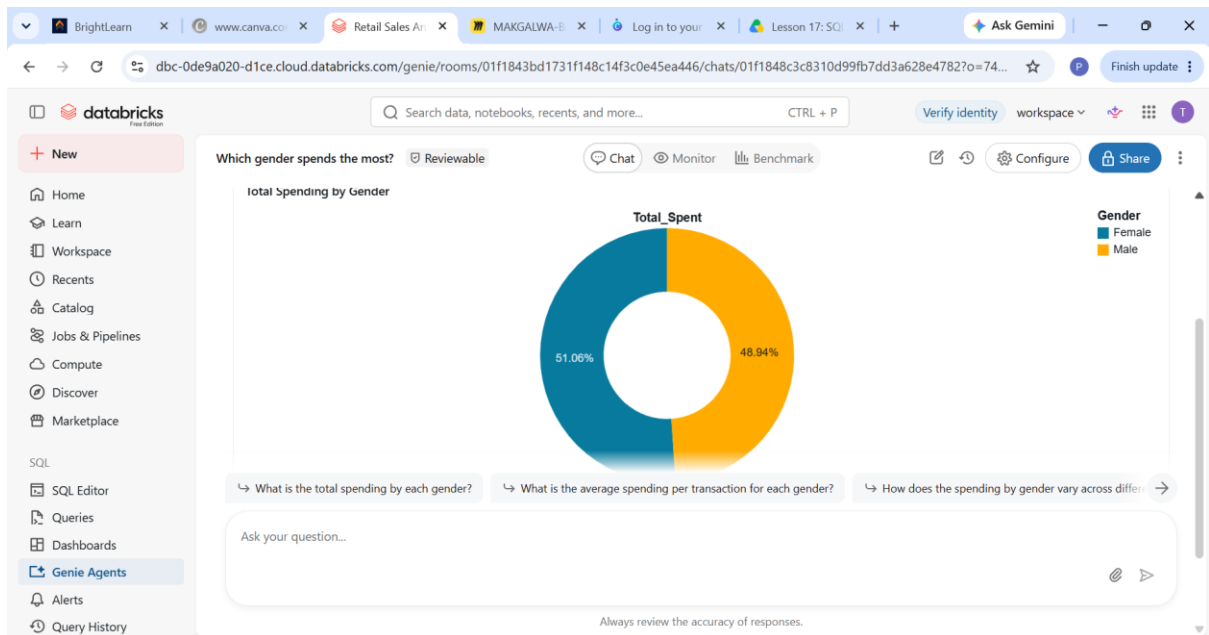
A few representative values for peak shopping hours include:

- 2023-05-23 00:00 with 29 sales
- 2023-07-14 00:00 with 26 sales
- 2023-05-16 00:00 with 26 sales
- 2023-02-17 00:00 with 21 sales
- 2023-02-05 00:00 with 20 sales

What are the peak shopping days of the week? What are the peak shopping months in the dataset? What is the total sales quantity by each day?

Ask your question...

Always review the accuracy of responses.



Which product category has the highest sales?

Analysis (click to view) sales_dataset

The Electronics product category has the highest sales, with a total sales amount of 156,905.

Result table (1 row)

Is this useful?

What are the total sales for each product category? Which product category has the highest quantity sold? What is the average price per unit for each product category?

Ask your question...

Always review the accuracy of responses.

Validations of 3 Answers

What is the total revenue generated for different categories of products??

```
1 SELECT 'Product Category',
2       SUM('Total Amount') AS Total_Revenue
3 FROM retail_sales.sales_schema.sales_dataset
4 GROUP BY 'Product Category'
5 ORDER BY total_revenue;
```

Parameters

Usage Guidance

Preview

	Product Category	Total_Revenue
1	Beauty	143,515
2	Clothing	155,580
3	Electronics	156,905

Browser tabs: BrightLearn, www.canva.co, Retail Sales An, MARGALWA-B, Log in to your, Lesson 17: SQL, Ask Gemini

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About Sources Instructions **Examples**

← What is the Age group are majority of the customers?

```
1 SELECT
2 CASE
3 WHEN Age BETWEEN 18 AND 25 THEN 'Young Adults'
4 WHEN Age BETWEEN 26 AND 35 THEN 'Youth'
5 WHEN Age BETWEEN 36 AND 59 THEN 'Adults'
```

Parameters

Usage Guidance

Preview

	Age_Group	Number_Of_Customers
1	Adults	511
2	Youth	205
3	Young Adults	169

Browser tabs: BrightLearn, www.canva.co, Retail Sales An, MARGALWA-B, Log in to your, Lesson 17: SQL, Ask Gemini

URL: dbc-0de9a020-d1ce.cloud.databricks.com/genie/rooms/01f1843bd1731f148c14f3c0e45ea446/chats/01f1848cb8981c12a5948d4762597d5d?o=74...

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Which age group bought most of the f Reviewable

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Total_Products_Bought

Age_Group

- Adults
- Youth
- Young Adults
- Pensioner

Which product category is most popular among the Youth age group (26-35)?

How does the total amount spent vary across different age groups?

Ask your question...

Always review the accuracy of responses.

Browser tabs: BrightLearn, www.canva.co, Retail Sales Ar, MAKGALWA, Log in to your, Lesson 17: SQ, Ask Gemini

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About Sources Instructions **Examples**

Can you do Sales analysis for the Retail Sales dataset?

```
1 SELECT
2   Date AS 'Date',
3   Quantity AS 'Quantity',
4   'Price per unit' AS 'Price per Unit',
5   Quantity * 'Price per unit' AS 'Total Amount'
```

Parameters

Usage Guidance

Preview

	Date	Quantity	Price per Unit	Total Amount
1	2024-01-01	3	500	1500
2	2024-01-01	1	30	30
3	2023-12-31	2	25	50

Key insights

Browser tabs: BrightLearn, www.canva.co, Retail Sales Ar, MAKGALWA, Log in to your, Lesson 17: SQ, Ask Gemini

URL: dbc-0de9a020-d1ce.cloud.databricks.com/genie/rooms/01f1843bd1731f148c14f3c0e45ea446/chats/01f18483eb801cf6a4a45cbca91525c7?o=747...

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key insights of retail sales dataset Reviewable

Total Units Sold

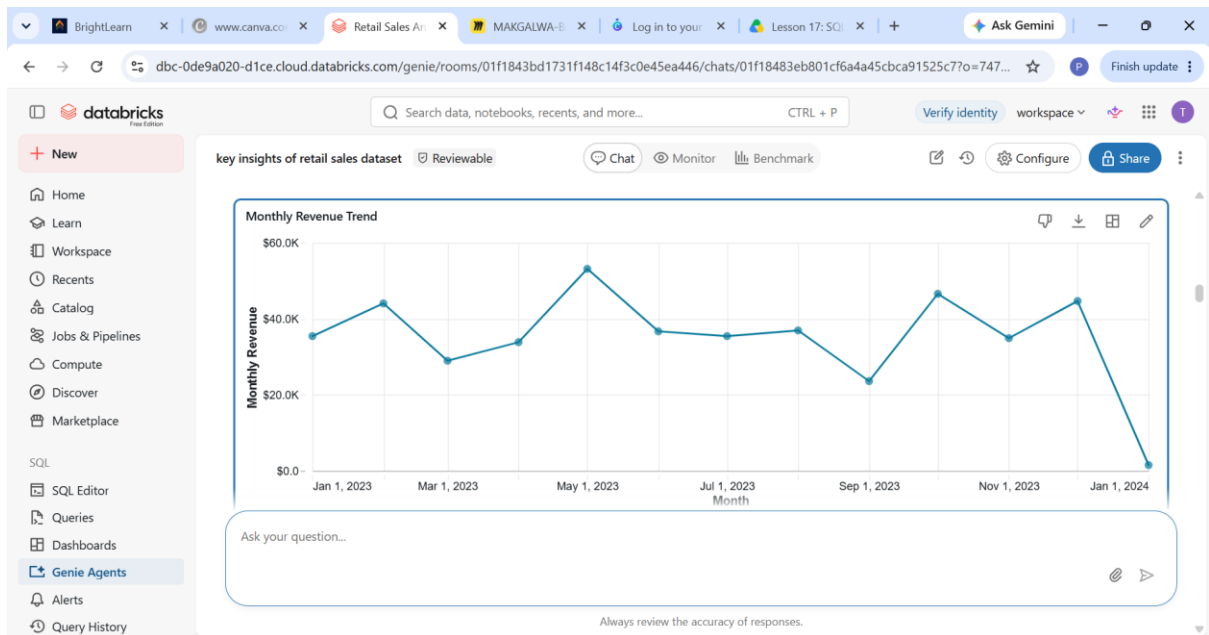
Product Category

- Clothing
- Electronics
- Beauty

Source: Units sold distribution by product category

Ask your question...

Always review the accuracy of responses.



The screenshot shows the Databricks interface with a summary table titled "Retail Sales Dataset: Key Insights". The table provides a high-level overview of the dataset, including the number of transactions, customers, total revenue, average transaction value, total units sold, and the first and last transaction dates.

Retail Sales Dataset: Key Insights						
The retail sales dataset covers 1,000 transactions from 1,000 unique customers between January 1, 2023, and January 1, 2024, generating \$456,000 in total revenue with an average transaction value of \$456.1						
Overall Sales Performance Summary						
Total Transactions	Total Customers	Total Revenue	Avg Transaction Value	Total Units Sold	First Transaction	Last Transaction
1,000	1,000	\$456,000	\$456.00	2,514	2023-01-01	2024-01-01

Business Recommendations

Retail sales company needs to run promotions, give discounts to products with popular items. If sales remain low, consider discontinuing them. Increase inventory for best-selling products. Create and facilitate loyal customer programs to increase customer retention. Focus marketing efforts on products that are selling more and generating higher revenues. Monitor weekly, monthly and quarterly sales trends. Identify and analyse slow moving product category and reduce its future purchase.

Conclusion

The Retail Sales Data Agent validates how conversational Artificial Intelligence is able to make retail data more accessible and actionable for business users. By allowing users to ask questions in natural language, the data agent retrieves and analyse retail sales data to provide meaningful insights without requiring advanced SQL analytical skills.

The use of key metrics such as customer demographics, sales trends, revenue of sales, and product category performance, the data agent assist in identifying business opportunities, and supporting data driven business decisions. It enables users to quickly answer questions such as which products generate the most revenue, and how sales change over time.

Beyond reporting historical data, the Retail Sales Data Agent also supports business growth by providing actionable recommendations, such as optimizing inventory, increasing customer retention, and focusing on high-performing products. These insights help retailers improve operational efficiency, maximizing of profits, and responding to changing customer demand.

Overall, the Retail Sales Data Agent demonstrates the value of transforming raw sales data into clear, actionable insights, empowering businesses to make faster, more informed, and data-driven decisions.

https://github.com/tsubielias-rgb/Retail_Sales_Data_Agent1.git